Yuan Jiang

Ph.D. Candidate, Industrial & Enterprise Systems Engineering
University of Illinois Urbana-Champaign
104 S. Mathews Ave., Urbana, IL 61801, USA
(+1) 217-693-8624 yuanj5@illinois.edu https://yuanjiang-lance.github.io Google Scholar

My research lies at the intersection of physics-informed AI and digital twins, developing principled, model-aware, and data-efficient frameworks for design, manufacturing, and lifecycle management of complex engineering systems. I integrate governing physics, multiphysics and multi-fidelity modeling, machine learning, optimization, and real-world observations to enable computationally efficient engineering design and optimization, adaptive state estimation and model updating, and uncertainty-aware lifecycle decision-making, while extending these capabilities to discover missing physics and transferable knowledge from model–reality discrepancies. My work spans advanced manufacturing, energy systems, and aerospace propulsion.

Education

University of Illinois Urbana-Champaign — **Urbana, IL, USA**
Ph.D., Industrial Engineering (Minor: Statistics) — Aug 2023 – May 2027 (Expected)
Advisor: Prof. Pingfeng Wang
Dissertation: Physics-informed multi-fidelity digital twins for design, diagnosis, and prognosis of complex engineering systems

Tongji University — **Shanghai, China**
M.Sc., Vehicle Engineering — Sep 2020 – Jun 2023
Advisor: Prof. Gang Niu
Thesis: Time-frequency feature extraction and health monitoring of rail short-pitch damage

Tongji University — **Shanghai, China**
B.Sc., Vehicle Engineering — Sep 2015 – Jul 2020

Industrial Experience

Rivian Automotive, Inc. — **Champaign, IL, USA**
Computational Methods Development Intern — May – Aug 2026
Developed a unified optimization framework for electric motor design, integrating heuristic and gradient-based optimization solvers with trust-region surrogate modeling strategies to reduce computational cost while maintaining design fidelity

NIO Automobile, Inc. — **Shanghai, China**
Power Systems Algorithm Intern — May – Aug 2023
Developed data-driven condition monitoring and fault detection methods for EV power swap and charging systems, translating large-scale operational and sensor data into real-time health indicators and early-warning signals

HiRain Technologies — **Shanghai, China**
Autonomous Driving Development Intern — Jun – Aug 2022
Contributed to perception software development and data processing pipelines for multi-camera autonomous driving systems, with exposure to large-scale, high-throughput data integration and real-time system constraints

NIO Automobile, Inc. — **Shanghai, China**
Battery Testing Intern — Feb – May 2021
Conducted experimental testing and data analysis for lithium-ion batteries, supporting model validation and data-informed performance assessment under real-world operating conditions

Awards & Honors

Research Awards

1. **ASME DAC Johnson Controls Best Paper Award**, *Design Automation Committee (DAC), ASME* 2026
2. **Doctoral Symposium Outstanding Presentation Award**, *Prognostics and Health Management Society* 2026
3. **IISE QCRE Best Student Poster Award**, *IISE Annual Conference & Expo* 2026
4. **IISE M&D Best Track Paper Finalist**, *IISE Annual Conference & Expo* 2026
5. **Best Poster Award (Second Place)**, *ISE Student Conference, UIUC* 2026
6. **ASME DAC Paper of Distinction**, *Design Automation Committee (DAC), ASME* 2025
7. **ASME IMECE Best Student Paper Award**, *ASME* 2024
8. **Outstanding Master's Thesis Award**, *Tongji University* 2023

9. **IEEE PHM Best Paper Award**, *IEEE Reliability Society* 2022

Fellowships, Scholarships & Academic Honors

1. **Mavis Future Faculty Fellow**, *University of Illinois Urbana-Champaign* 2026
2. **Jerry S. Dobrovolny Fellowship**, *University of Illinois Urbana-Champaign* 2023
3. **Tongji Outstanding Graduate**, *Tongji University* 2023
4. **National Scholarship of China**, *Ministry of Education of the People's Republic of China* 2019, 2022
5. **Shanghai Outstanding Graduate**, *Shanghai Municipal Education Commission* 2020
6. **First-Class Scholarship**, *Tongji University* 2016, 2017, 2018

Professional Recognition

1. **IOE-ISyE-MS&E Joint Rising Star**, *IOE (UMich), ISyE (Georgia Tech), MS&E (Stanford)* 2026
2. **Doctoral Symposium Participant Award**, *Prognostics and Health Management Society* 2026
3. **IEEE TIM Outstanding Reviewer**, *IEEE Instrumentation and Measurement Society* 2025

Research Experience

Physics-Guided Multi-Fidelity Digital Twins for Hydraulic Drivetrain Remanufacturing Diagnostics
Sponsor: Department of Energy (DOE) REMADE Institute, National Science Foundation (NSF) Jan 2026 – Present
Collaboration: University of Connecticut, John Deere

- **Multi-Fidelity Simulation:** Developed fluid-structure interaction models in COMSOL and a reduced-fidelity strategy replacing explicit contact with spring-damper approximations, cutting simulation time to 1/4 while preserving transient features
- **Physics-Guided Neural Operator:** Proposed a Fourier-embedded neural operator with physics-guided multi-fidelity training for transient pressure prediction under drivetrain defects, achieving 1% prediction error, 10% lower than baseline models
- **Digital-Twin-Based Diagnosis:** Developed inverse diagnosis using condition-labeled digital-twin libraries and similarity matching, achieving 98% diagnostic accuracy within 1 s; mentored a junior Ph.D. student on noise-robust diagnosis
- **Outcomes & Impact:** Co-first-authored **1 journal paper** under second-round review in ASME JMD and **1 conference paper** accepted at ASME IDETC-CIE 2026; extending the framework toward trustworthy diagnosis and post-remanufacturing performance prediction

Physics-Informed Machine Learning for Large-Scale Battery Thermal-Health Management and Design
Sponsor: National Science Foundation (NSF) Jan 2024 – Present
Collaboration: Stanford University, UIUC Mechanical Engineering, John Deere, Ford

- **Multi-Fidelity Physics-Informed Learning:** Proposed MFPI-CNN combining physics-informed pre-training with limited-data fine-tuning, achieving 2% prediction error with 5$\times$ fewer labels; integrated with optimization to improve battery temperature uniformity by 7% with 5$\times$ speedup compared with simulation-based method
- **Physics-Informed Neural Operator:** Proposed mesh-free neural operators (PI-CAON and PI-SCAON) combining Fourier features, hard-constrained boundaries, and attention mechanisms for inter-cell thermal interactions, achieving $<$2% error on 20-cell packs with data-free training and 250$\times$ faster inference
- **Multi-modal Sensing:** Developed a physics-based framework for cell-current reconstruction from magnetic measurements with POD-based sensor placement optimization, reducing measurement points from 9 to 4 among 500 candidates
- **Outcomes & Impact:** Published **2 journal papers** in RESS and EAAI, and **3 conference papers**; received **ASME DAC Best Paper Award** and was named an **IISE M&D Best Track Paper Finalist**

Physics-Informed Adaptive Digital Twin for Hall Thruster Erosion Degradation Modeling and Prognostics
Sponsor: Air Force Office of Scientific Research (AFOSR) Sep 2023 – Present
Collaboration: UIUC Aerospace Engineering, Georgia Institute of Technology

- **Degradation Mechanism Discovery:** Built 2D particle-in-cell and 1D fluid-based plasma-erosion models with multi-fidelity calibration, improving erosion prediction by 10% and accelerating lifecycle simulation from 2 weeks to 6 hours; identified thrust as an in-situ degradation indicator
- **Adaptive Digital Twin:** Proposed an adaptive self-cognizant dynamic system integrating mechanism-informed state-space modeling, neural networks, and particle filtering for online state and model updates, achieving $<$5% health estimation error with 0.14-s updates
- **Uncertainty-Aware Prognostics:** Developed remaining useful life prediction under manufacturing and environmental uncertainty, achieving $<$10% error over a 5000-h thruster lifecycle and improving prognostic accuracy by $>$3% over baselines
- **Outcomes & Impact:** Published **1 invited journal paper** in ASME JMD and **2 peer-reviewed conference papers**; received **ASME DAC Paper of Distinction** and **IISE QCRE Best Student Poster Award**

Robust Signal Processing for Structural and Mechatronic System Health Monitoring
Sponsor: National Science Foundation (NSF) Sep 2023 – Jul 2025

- **Mechatronic Fault Diagnosis:** Developed an iterative adaptive Vold-Kalman filter (IAVKF) with frequency refinement and adaptive bandwidth selection for mechatronic transmission diagnosis under variable-speed conditions, improving instantaneous-frequency estimation accuracy by $>$10 dB
- **Structural Health Monitoring:** Developed an iterative dispersive Vold-Kalman filter (IDVKF) with joint decomposition and local adaptive bandwidths for overlapping dispersive signals with frequency-varying bandwidths in structural health monitoring, improving denoising performance by 2–10 dB
- **Outcomes & Impact:** Published **2 first-authored journal papers** in IEEE TII

Publications

[#] indicates equal contribution

11 published peer-reviewed journal papers (6 first-author), including IEEE TII, RESS, MSSP, EAAI, and ASME JMD.

Journal Papers

[J1] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, and Pingfeng Wang. Physics-informed cross-attention operator network with hard-constrained Fourier features for heat map prediction of large-scale battery packs. *Engineering Applications of Artificial Intelligence*, 176 (2026) 114833. DOI: 10.1016/j.engappai.2026.114833.

- ISE Student Conference Best Poster Award Second Place (2nd out of 28)

[J2] **Yuan Jiang**, Yiyue Jiang, Zheng Liu, Yuejian Chen, and Pingfeng Wang. Iterative dispersive Vold-Kalman filter with local adaptive bandwidth for dispersive signal decomposition in structural health monitoring. *IEEE Transactions on Industrial Informatics*, (2026). DOI: 10.1109/TII.2026.3702939.

[J3] Yiyue Jiang, **Yuan Jiang**, and Pingfeng Wang. A review on mitigating thermal runaway propagation in battery packs: From mechanisms to modeling and design optimization. *Energy Advances*, 5(6) (2026) 777-809. DOI: 10.1039/D5YA00351B.

[J4] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang. Remaining useful life prediction for Hall thrusters based on adaptive self-cognizant dynamic system and multi-physics modeling. *Journal of Mechanical Design*, 148(7) (2026) 071701. DOI: 10.1115/1.4070967.

- ASME DAC Paper of Distinction (Top 10 out of 103)
- Invited to JMD Special Issue: Selected Papers from IDETC 2025

[J5] Zheng Liu, Yanwen Xu, **Yuan Jiang**, Anabel Renteria, Parth Bansal, Chenlong Xu, Pingfeng Wang, and Yumeng Li. Uncertainty quantification of additively manufactured architected cellular materials for energy absorption applications. *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering*, 11(3) (2025). DOI: 10.1115/1.4066933.

[J6] Pouya Kabirzadeh, Zheng Liu, Mostafa Olyaei, Haoyun Qiu, Yashraj Gurumukhi, Harsh Tyagi, **Yuan Jiang**, Vivek S. Garimella, Bakhshish Preet Singh, Yumeng Li, Pingfeng Wang, and Nenad Miljkovic. Integrating heat transfer and control optimization: A comprehensive review of battery thermal management systems. *Journal of Energy Storage*, 131 (2025) 117289. DOI: 10.1016/j.est.2025.117289.

[J7] **Yuan Jiang**[#], Zheng Liu[#], Pouya Kabirzadeh, Yulun Wu, Yumeng Li, Nenad Miljkovic, and Pingfeng Wang. Multi-fidelity physics-informed convolutional neural network for heat map prediction of battery packs. *Reliability Engineering & System Safety*, 256 (2025) 110752. DOI: 10.1016/j.ress.2024.110752.

[J8] Yuejian Chen, Zihan Li, **Yuan Jiang**, Chunsheng Yang, Min Xia, and Ke Feng. Enhanced sparse LPV-ARMA model with ensemble basis functions for mechatronic transmission fault detection under variable speed conditions. *IEEE Internet of Things Journal*, 12(11) (2025) 17223-17232. DOI: 10.1109/JIOT.2025.3535580.

[J9] Yuejian Chen, Zihan Li, **Yuan Jiang**, Dao Gong, and Kai Zhou. Sparse LPV-ARMA model for non-stationary vibration representation and its application on gearbox tooth crack detection under variable speed conditions. *Mechanical Systems and Signal Processing*, 224 (2025) 112161. DOI: 10.1016/j.ymssp.2024.112161.

[J10] **Yuan Jiang**, Yuejian Chen, and Pingfeng Wang. An iterative adaptive Vold-Kalman filter for non-stationary signal decomposition in mechatronic transmission fault diagnosis under variable speed conditions. *IEEE Transactions on Industrial Informatics*, 20(8) (2024) 10510-10519. DOI: 10.1109/TII.2024.3393536.

[J11] **Yuan Jiang** and Gang Niu. An iterative frequency-domain envelope-tracking filter for dispersive signal decomposition in structural health monitoring. *Mechanical Systems and Signal Processing*, 179 (2022) 109329. DOI: 10.1016/j.ymssp.2022.109329.

Journal Papers under Review

[J12] **Yuan Jiang**[#], Parth Bansal[#], Mohammad Mundiwala, Chao Hu, and Pingfeng Wang. Physics-guided multi-fidelity neural operator digital twin for wet clutch hydraulic pressure prediction. *Journal of Mechanical Design.* (Under second-round review)

Conference Proceedings

[C1] **Yuan Jiang** and Pingfeng Wang. Physics-informed self-cross attention operator network for heat map prediction in large-scale battery thermal management systems. *ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2026)*, Houston, TX, USA. 24 Aug 2026, 10 pages. (Accepted & Presented)

- ASME DAC Johnson Controls Best Paper Award (1st out of 107)

[C2] Jiayi Zhao[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang. A digital-twin-based framework for inverse diagnosis of wet clutch hydraulic pressure signals using time-series similarity matching. *ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2026)*, Houston, TX, USA. 24 Aug 2026, 10 pages. (Accepted & Presented)

[C3] Yulun Wu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang. From local to global: CNN surrogates for scaling layout-resolved temperature prediction in battery thermal management. *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2026)*, Houston, TX, USA. 24 Aug 2026, 10 pages. (Accepted & Presented)

[C4] **Yuan Jiang** and Pingfeng Wang. Multi-fidelity adaptive self-cognizant dynamic system for remaining useful life prediction of Hall thrusters. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 pages. (Accepted and Presented)

- IISE QCRE Best Student Poster Award (1st out of 18)

[C5] Yiyue Jiang[#], **Yuan Jiang**[#], and Pingfeng Wang. Spatiotemporal design-aware surrogate modeling of thermal runaway propagation in Li-ion battery packs. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 pages. (Accepted and Presented)

- IISE M&D Best Track Paper Finalist (Top 4 out of 40)

[C6] Yanbin Luo, **Yuan Jiang**, and Pingfeng Wang. Degradation behavior and health monitoring of GaN HEMTs under high-temperature stress conditions. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 pages. (Accepted and Presented)

[C7] Jiayi Zhao, **Yuan Jiang**, Yiyue Jiang, and Pingfeng Wang. A complexity-aligned benchmark for adaptive sampling in GP global fitting. *IISE Annual Conference & Expo (IISE 2026)*, Arlington, TX, USA. 17 May 2026, 6 pages. (Accepted and Presented)

[C8] Alexandra N. Leeming, **Yuan Jiang**, Stephen Messing, Siyuan Wang, Pingfeng Wang, J. Gary Eden, Peter D. Dragic, Joshua L. Rovey, Valentin Korman, and Kurt Polzin, Exploration of an in-situ Hall thruster erosion and plasma diagnostic sensor. *AIAA Science and Technology Forum and Exposition (AIAA SciTech 2026)*, Orlando, FL, USA. 12 Jan 2026, 14 pages. DOI: 10.2514/6.2026-0929.

[C9] Zheng Liu[#], **Yuan Jiang**[#], Yumeng Li, and Pingfeng Wang, Physics-informed machine learning enhanced battery pack optimization. *IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+EATS 2025)*, Anaheim, CA, USA. 18 June 2025, 5 pages. DOI: 10.1109/ITEC63604.2025.11098093.

[C10] **Yuan Jiang**, Alexandra N. Leeming, Joshua L. Rovey, and Pingfeng Wang, Prognostics of Hall thruster erosion using multiphysics-based modeling and machine learning. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan 2025, 7 pages. DOI: 10.1109/RAMS48127.2025.10935282.

[C11] Zheng Liu, **Yuan Jiang**, Yumeng Li, and Pingfeng Wang, Physics-informed machine learning for battery pack thermal management. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan 2025, 7 pages. DOI: 10.1109/RAMS48127.2025.10935157.

[C12] Shimeng Yang, **Yuan Jiang**, Zheng Liu, and Pingfeng Wang, Transductive transfer learning features for prognostics and health management. *Annual Reliability and Maintainability Symposium (RAMS 2025)*, Destin, FL, USA. 27 Jan 2025, 7 pages. DOI: 10.1109/RAMS48127.2025.10935143.

[C13] Parth Bansal, **Yuan Jiang**, Zhou Li, Sergio Cordero, Zahra Heussen, Debbie Senesky, Pingfeng Wang, and Yumeng Li, Multiphysics modeling and simulation of gas sensor for NO_2 detection. *ASME International Mechanical Engineering Congress and Exposition (IMECE 2024)*, Portland, OR, USA, 17 Nov 2024, 6 pages. DOI: 10.1115/IMECE2024-145663.

- ASME IMECE Best Student Paper Award

[C14] **Yuan Jiang** and Gang Niu, Rail local damage detection based on recursive frequency-domain envelope tracking filter and rail impact index. *IEEE Global Reliability and Prognostics and Health Management Conference (PHM 2022)*, Yantai, China. 13 Oct 2022, 7 pages. DOI: 10.1109/PHM-Yantai55411.2022.9942082.

- IEEE PHM Best Paper Award (Top 5 out of 274)

[C15] Hongyang Zhao, **Yuan Jiang**, and Gang Niu, A tacholess order tracking method based on extended intrinsic chirp component decomposition for gears under large speed variation conditions. *Journal of Physics: Conference Series*, 2184(1)

(2022) 12052. DOI: 10.1088/1742-6596/2184/1/012052.

[C16] **Yuan Jiang**, Hongyang Zhao, and Gang Niu, Intelligent rolling bearing fault diagnosis under variable speed conditions without tachometers. *IEEE Global Reliability and Prognostics and Health Management Conference (PHM 2021)*, Nanjing, China. 15 Oct 2021, 7 pages. DOI: 10.1109/PHM-Nanjing52125.2021.9613049.

Presentations

[T1] Physics-informed multi-fidelity digital twins for prognostics and health management of complex engineering systems
- Doctoral symposium, *PHM Society Conference 2026*, Charlotte, NC, USA, 28 Sep 2026
- PHM Doctoral Symposium Outstanding Presentation Award (Top 1 out of 10)

[T2] Physics-informed self-cross attention operator network for heat map prediction in large-scale battery thermal management systems
- Conference presentation, *IDETC-CIE 2026*, Houston, TX, USA, 24 Aug 2026

[T3] Multi-fidelity adaptive self-cognizant dynamic system for remaining useful life prediction of Hall thrusters
- Conference presentation, *IISE 2026*, Arlington, TX, USA, 17 May 2026

[T4] Degradation behavior and health monitoring of GaN HEMTs under high-temperature stress conditions
- Conference presentation, *IISE 2026*, Arlington, TX, USA, 17 May 2026

[T5] Remaining useful life prediction for Hall thrusters based on adaptive self-cognizant dynamic system and multi-physics modeling
- Conference presentation, *IDETC-CIE 2025*, Anaheim, CA, USA, 18 Aug 2025

[T6] Prognostics of Hall thruster erosion using multiphysics-based modeling and machine learning
- Conference presentation, *RAMS 2025*, Destin, FL, USA, 27 Jan 2025

[T7] Rail local damage detection based on recursive frequency-domain envelope tracking filter and rail impact index
- Conference presentation, *PHM 2022*, Yantai, China, 13 Oct 2022

[T8] Intelligent rolling bearing fault diagnosis under variable speed conditions without tachometers
- Conference presentation, *PHM 2021*, Nanjing, China, 15 Oct 2021

Teaching & Mentoring Experience

Teaching Experience

SE 101: Engineering Graphics & Design — **University of Illinois Urbana-Champaign**
Lab Instructor: Undergraduate Required (2 Lab Sections, 30 Students) — Fall 2026

- Instructed freehand engineering sketching, including one- and two-point perspective, isometric and oblique sketches, multi-view and section views, auxiliary views, and dimensioning and tolerancing
- Instructed 3D CAD modeling using Autodesk Fusion 360, including part modeling, patterning, drawings, advanced modeling techniques, assemblies, freeform modeling, rendering, and animation
- Guided team design projects involving product selection, hands-on dissection and measurement, task decomposition, and 3D reconstruction; mentored students on challenging modeling problems and evaluated final reports and presentations

SE 311: Engineering Design Analysis — **University of Illinois Urbana-Champaign**
Teaching Assistant: Undergraduate Required (100 Students) — Spring 2026

- Delivered guest lectures on stress analysis, covering state of stress, stress transformation, principal stresses, and maximum shear stress, with emphasis on engineering applications and analytical problem solving
- Delivered exam review lectures covering elasticity, stress analysis, energy methods, and introductory finite element analysis, reinforcing systematic approaches to solving engineering mechanics problems
- Developed and delivered a virtual Abaqus tutorial for finite element analysis, guiding students through finite element modeling, post-processing, and project completion involving stress analysis and validation against analytical solutions

SE 411: Reliability Engineering — **University of Illinois Urbana-Champaign**
Teaching Assistant: Undergraduate/Graduate Elective (30 Students) — Spring 2025

- Delivered guest lectures on reliability analysis and reliability-based design optimization (RBDO), covering problem formulation, algorithmic implementation, and engineering interpretation
- Designed an RBDO course project and mentored students on modeling, optimization algorithms, and coding implementation
- Led exam review sessions with structured problem-solving tutorials to reinforce core concepts and analytical skills

SE 450: Decision Analysis — **University of Illinois Urbana-Champaign**
Teaching Assistant: Undergraduate/Graduate Elective (45 Students) — Fall 2024, Fall 2025

- Contributed to instruction by leading discussion sections and providing academic support via office hours
- Delivered exam review lectures covering multivariate decision-making, decision trees, and uncertainty-based decision-making, with an emphasis on systematic problem formulation and solution strategies

Rail Vehicle Systems Health Management **Tongji University**
Teaching Assistant: Undergraduate Required (60 Students, Instructed in English) Spring 2022

- Supported course delivery by assisting lecturers and coordinating laboratory components
- Led laboratory sessions on bearing and gearbox test rigs, covering experimental setup, data acquisition, and signal processing
- Instructed students on vibration-based fault diagnosis using FFT, wavelet transforms, and machine learning methods

Student Mentoring

Jiayi Zhao *(Ph.D. Student, UIUC)* **Jan 2026 – Present**
Digital-Twin-Based Inverse Diagnosis for Hydraulic Drivetrain Remanufacturing

- Primary technical mentor for research methodology, guiding digital-twin framework design, similarity-based diagnostic approach, and trustworthy decision-making methods
- Co-first-authored ASME IDETC-CIE 2026 conference paper (accepted); journal paper on trustworthy diagnosis in preparation

Jingsen Yang *(M.S., UIUC)* **Nov 2025 – Present**
Multi-fidelity Surrogate Modeling and Data Fusion based on Foundational Tabular Model

- Primary technical mentor for surrogate modeling, data processing, and methodological development
- Thesis and conference paper in preparation

Yanbin Luo *(M.S., UIUC)* **Jul 2025 – May 2026**
Experimental Degradation Modeling of High-Electron-Mobility Transistors (HEMTs) under Extreme Operating Conditions

- Primary technical mentor for experimental design, data processing, and degradation modeling
- Guided master's thesis and IISE conference paper preparation (accepted and presented)

Academic Service

Conference Leadership

- Session Chair, Physics-Informed, Hybrid, and Model-Based PHM, *PHM Society Conference* 2026
- Session Chair, Prognostics and Reliability in Power and Energy Systems, *IISE Annual Conference & Expo* 2026

Departmental, University & Community Service

- IISE QCRE Student Board Member 2026 – Present
- ISE Student Conference Organizing Committee, *Industrial & Enterprise Systems Engineering, UIUC* 2026 – 2027
- ISE Graduate Program Ambassador, *Industrial & Enterprise Systems Engineering, UIUC* 2026 – Present

Journal Reviewer

- Reliability Engineering & System Safety
- Engineering Applications of Artificial Intelligence
- Expert Systems with Applications
- IEEE Transactions on Reliability
- Neurocomputing
- Journal of Process Control
- Journal of Vibration and Control
- Scientific Reports
- IEEE Transactions on Industrial Informatics
- Mechanical Systems and Signal Processing
- IEEE Transactions on Instrumentation & Measurement
- ISA Transactions
- IEEE Sensors Journal
- Computers & Structures
- Microelectronics Journal
- Journal of Rail and Rapid Transit

Conference Reviewer

- ASME International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE)

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- American Society of Mechanical Engineers (ASME)
- Institute of Industrial and Systems Engineers (IISE)
- Institute for Operations Research and the Management Sciences (INFORMS)